Qingzhao Zhang

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RESEARCH INTERESTS

System security (e.g., cyber-physical systems, autonomous driving), software security (e.g., program analysis, formal verification), AI security (e.g., adversarial attacks, robustness).

EDUCATION

University of Michigan *Sep 2019 – May 2025*
***Ph.D.** in Computer Science and Engineering*

University of Michigan *Sep 2019 – Dec 2023*
***M.S.** in Computer Science and Engineering*

Shanghai Jiao Tong University *Sep 2015 – May 2019*
***B.E.** in Computer Engineering*

WORK EXPERIENCE

Assistant Professor | University of Arizona, ECE *Aug 2025 - Present*

- Lead *Secure Autonomous Systems* (SAS) research group working on computer security and systems. We advance the security, resilience, and trustworthiness of emerging autonomous systems (e.g., self-driving vehicles, drones, and robots) through cross-layer design and analysis.

Research Assistant/Research Fellow | University of Michigan, EECS *May 2020 - Aug 2025*
Supervisor: Prof. Z. Morley Mao

- Research on software/system for cyber-physical system safety — *AVChecker*, the first traffic rule compliance checker on autonomous driving software; *SmtConf*, safety vetting of industrial control system configuration.
- Research on adversarial machine learning on cyber-physical systems — Adversarial attack and mitigation on trajectory prediction on autonomous driving; Analyzed data fabrication vulnerability on collaborative perception.
- Research on robustness of vehicular network — *RAO*, collaborative perception with asynchronous sensors.
- Research on robust perception algorithms of autonomous driving, and large language model security/efficiency.

Software Engineer Intern | Google *May 2023 - July 2023*

- Designed and implemented formal verification solutions to enhance the security properties of an embedded system kernel (based on open-sourced Tock OS), involving modular model checking and theorem proving.

Software Engineer Intern | Google *May 2022 - July 2022*

- Designed and implemented static analysis checks based on Android Lint for Google's Android tests, which was deployed to assist Google developers to write high-quality unit tests.

PUBLICATIONS

Conference Papers

- **[S&P '26]** Jiarui Li, Joey Brewington, **Qingzhao Zhang**, Z. Morley Mao. "Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection", *The 47th IEEE Symposium on Security and Privacy*.
- **[COLM '25]** Shuowei Jin, Yongji Wu, Haizhong Zheng, **Qingzhao Zhang**, Matthew Lentz, Z Morley Mao, Atul Prakash, Feng Qian, Danyang Zhuo. "Plato: Plan to Efficiently Decode for Large Language Model Inference", *Conference on Language Modeling*.

- **[ICML '25]** Shuowei Jin*, Xueshen Liu*, **Qingzhao Zhang**, Z. Morley Mao. "Compute Or Load KV Cache? Why Not Both?", *Forty-Second International Conference on Machine Learning*.
- **[ICLR '25]** Minkyoung Cho, Yulong Cao, Jiachen Sun, **Qingzhao Zhang**, Marco Pavone, Jeong Joon Park, Heng Yang, Z. Morley Mao. "Cocoon: Robust Multi-Modal Perception with Uncertainty-Aware Sensor Fusion", *The International Conference on Learning Representations*.
- **[USENIX Security '24] Qingzhao Zhang**, Shuowei Jin, Ruiyang Zhu, Jiachen Sun, Xumiao Zhang, Qi Alfred Chen, Z. Morley Mao. "On Data Fabrication in Collaborative Vehicular Perception: Attacks and Countermeasures", *Proceedings of the 33nd USENIX Security Symposium*.
- **[ICLR '24]** Jiachen Sun, Haizhong Zheng, **Qingzhao Zhang**, Atul Prakash, Z. Morley Mao, Chaowei Xiao. "CALICO: Self-Supervised Camera-LiDAR Contrastive Pre-training for BEV Perception", *The International Conference on Learning Representations*.
- **[MobiCom '23] Qingzhao Zhang***, Xumiao Zhang*, Ruiyang Zhu*, Fan Bai, Mohammad Naserian, Z. Morley Mao. "Robust Real-time Multi-vehicle Collaboration on Asynchronous Sensors", *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*.
- **[RAID '22] Qingzhao Zhang**, Xiao Zhu, Mu Zhang, Z. Morley Mao. "Automated Runtime Mitigation for Misconfiguration Vulnerabilities in Industrial Control Systems", *Proceedings of the 25th International Symposium on Research in Attacks, Intrusions and Defenses*.
- **[CVPR '22] Qingzhao Zhang**, Shengtuo Hu, Jiachen Sun, Qi Alfred Chen, Z. Morley Mao. "On adversarial robustness of trajectory prediction for autonomous vehicles", *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- **[SIGMETRICS '21] Qingzhao Zhang**, David Ke Hong, Ze Zhang, Qi Alfred Chen, Scott Mahlke, Z Morley Mao. "A systematic framework to identify violations of scenario-dependent driving rules in autonomous vehicle software", *Proceedings of the ACM on Measurement and Analysis of Computing Systems*.

TEACHING AND MENTORSHIP

Course Instructor, University of Arizona

- Spring 2026, ECE 402C/502C Operating System Design.

PhD Students, University of Arizona

- Yutong Liu, started in Fall 2025.

Supervisor of Undergraduate Research, University of Michigan *2022*

- Multidisciplinary Design Program (MDP) at University of Michigan. Lead cybersecurity research and supervised a team of four students.

Research Mentorship, University of Michigan *2020-2024*

- Research on program analysis, autonomous driving, adversarial machine learning.
- Mentees: Charles Ziegenbein Jr., Kevin Zhang, Andrew Wei, Xingyu Wang, Jingjia Peng, Ziyang Xiong, Runting Zhang.

SERVICES

- Conference Committee Member: S&P 2027, AsiaCCS 2027, CCS 2026/2025
- Conference Reviewer: NeurIPS 2026, COLM 2026, CVPR 2026, ECCV 2026, ICLR 2026/2025, ICRA 2025, ACM MM 2024, IV 2024/2023
- Journal Reviewer: TDSC, IoT-J, Sensors, TIFS, ITSM
- Workshop Committee Member: RICSS 2024, RICSS 2025
- Artifact Reviewer: Usenix Security 2022
- Pre-review Task Force: NSDI 2025